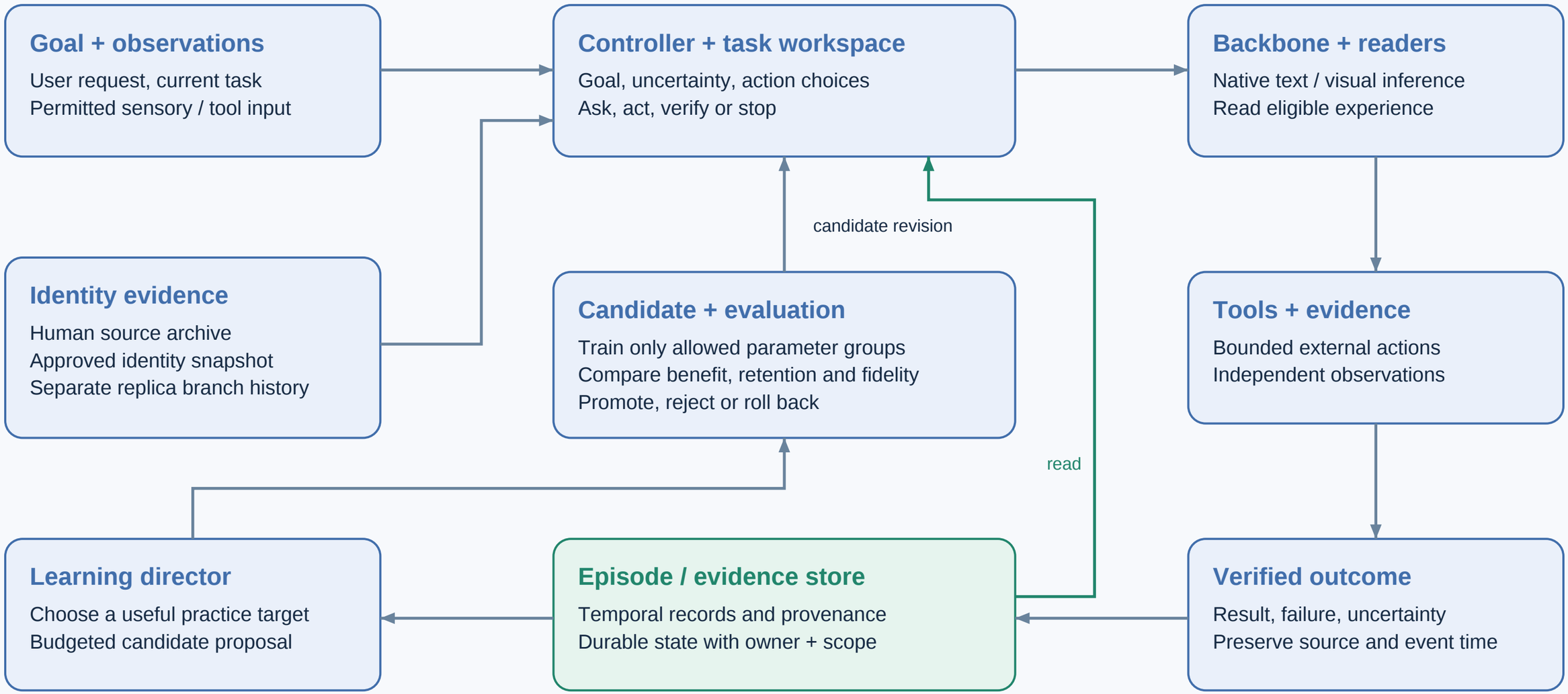
The complete experience-to-learning system

Stable foundation, evidence-backed memory, a bounded control loop and separately owned identity.



Source-person identity is evidence, not a self-generated biography. New agent experience belongs to the replica branch.

Shipped primitives / CPU checks

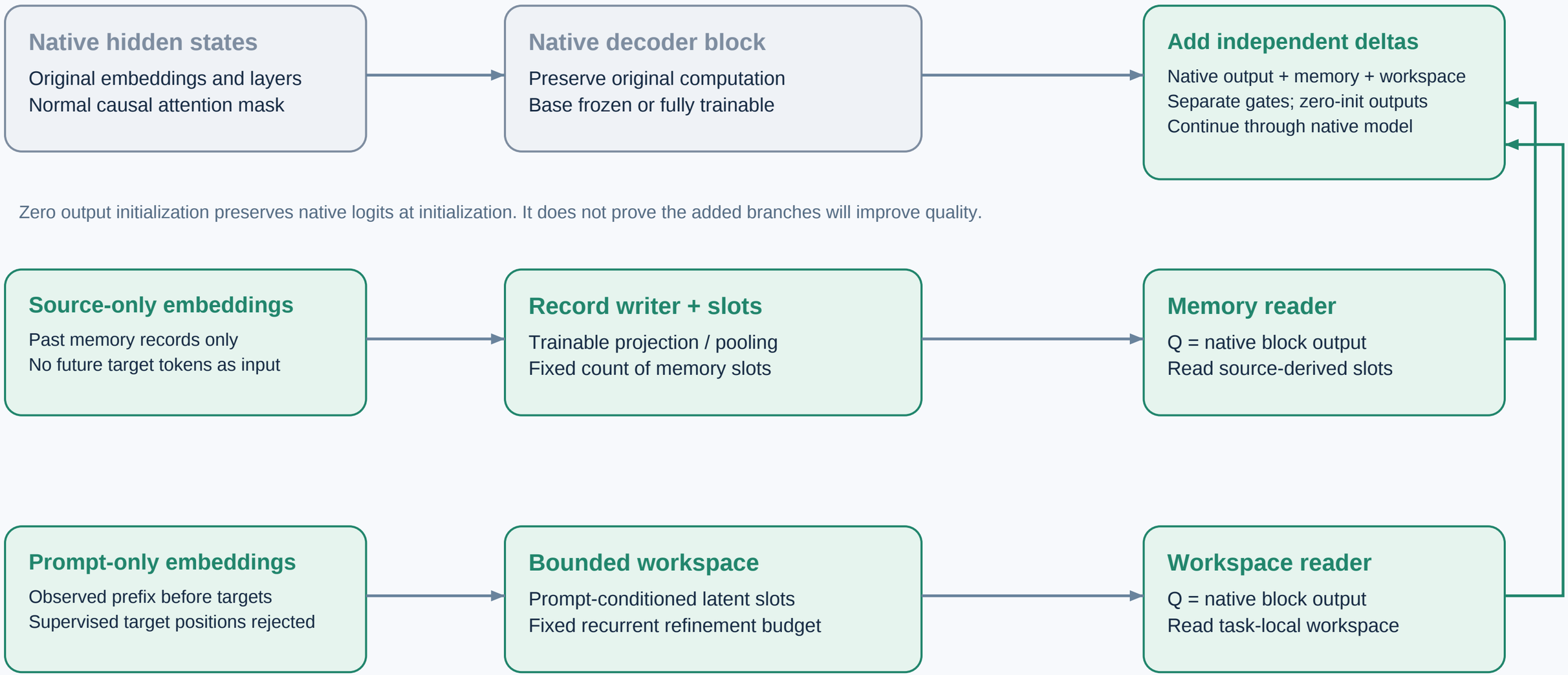
Proposed system integration

Optional research branch

Proposed research architecture; CPU contract tests are not model-quality evidence.

Small-model integration inside the native backbone

Shipped text scaffold: one native Llama block receives two independently gated residual additions.



Zero output initialization preserves native logits at initialization. It does not prove the added branches will improve quality.

One in-flight batch owns the context until backward completes. Save the native base and sidecars separately; cached generation is pending.

Shipped primitives / CPU checks

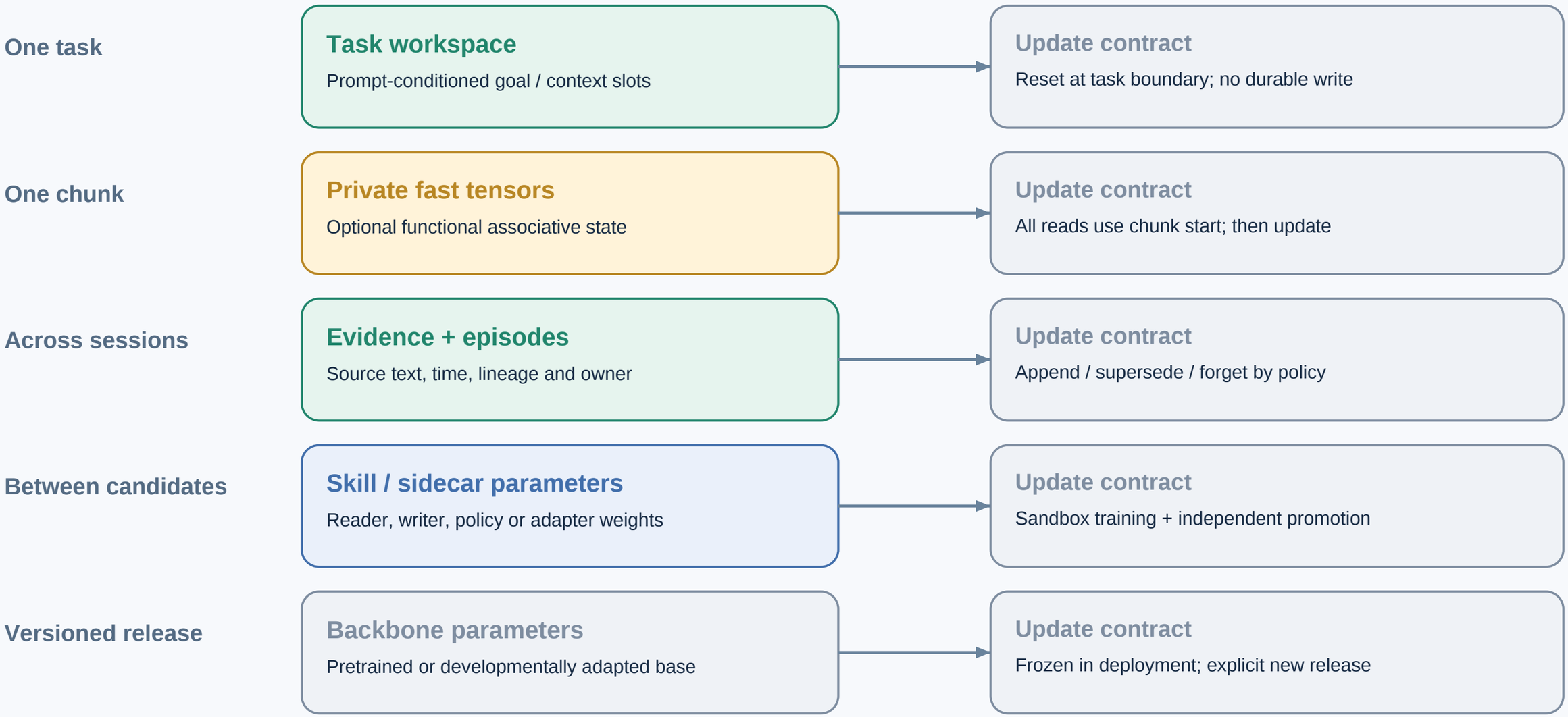
Proposed system integration

Optional research branch

Proposed research architecture; CPU contract tests are not model-quality evidence.

Different states learn on different timescales

Persistence, update authority and neural parameters are explicit rather than hidden inside one memory object.



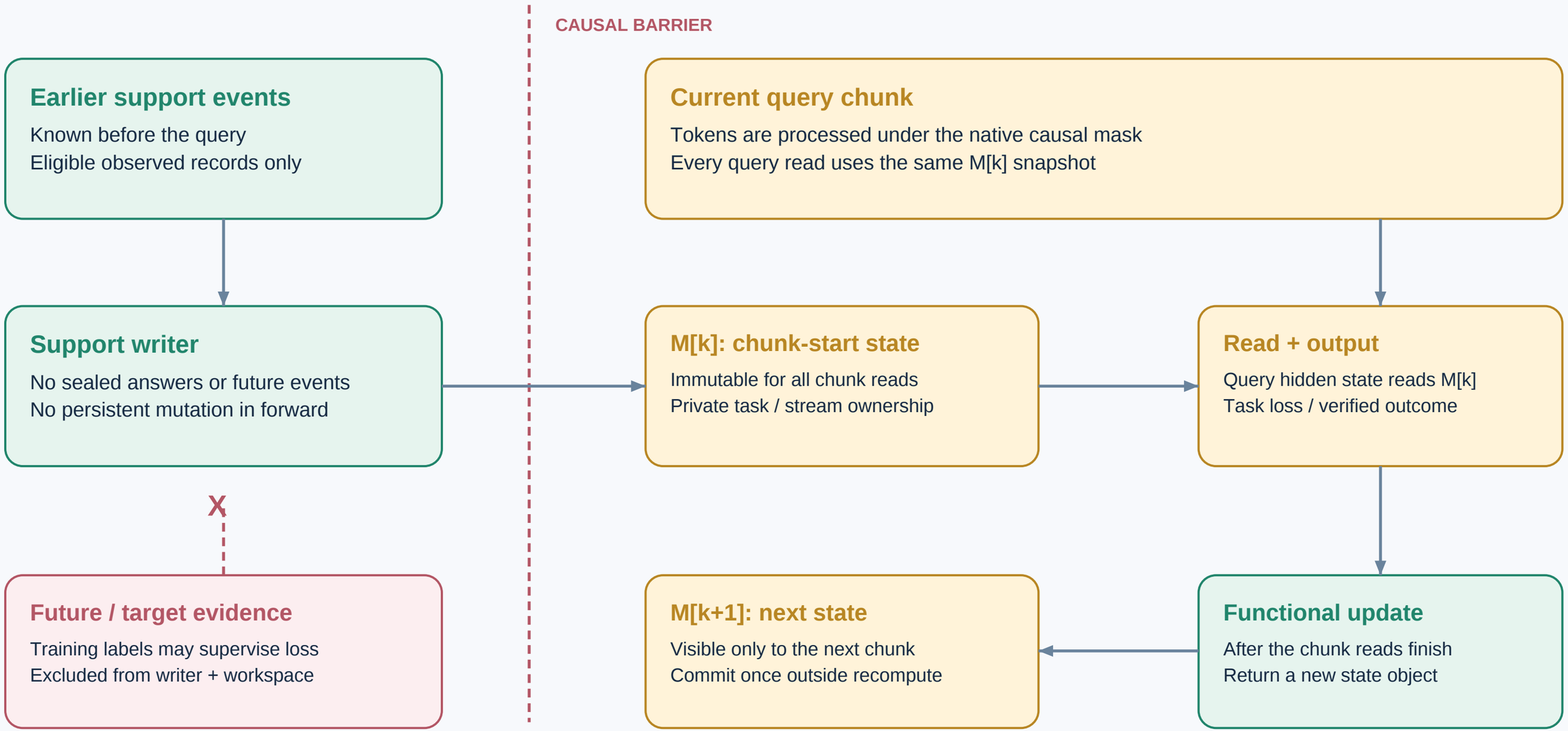
A stored fact, an online tensor update and a promoted parameter revision are different improvement claims.

Shipped primitives / CPU checks Proposed system integration Optional research branch

Proposed research architecture; CPU contract tests are not model-quality evidence.

Causal memory: support before query, commit after read

A shared immutable chunk-start state prevents one query token from seeing a later write in the same chunk.



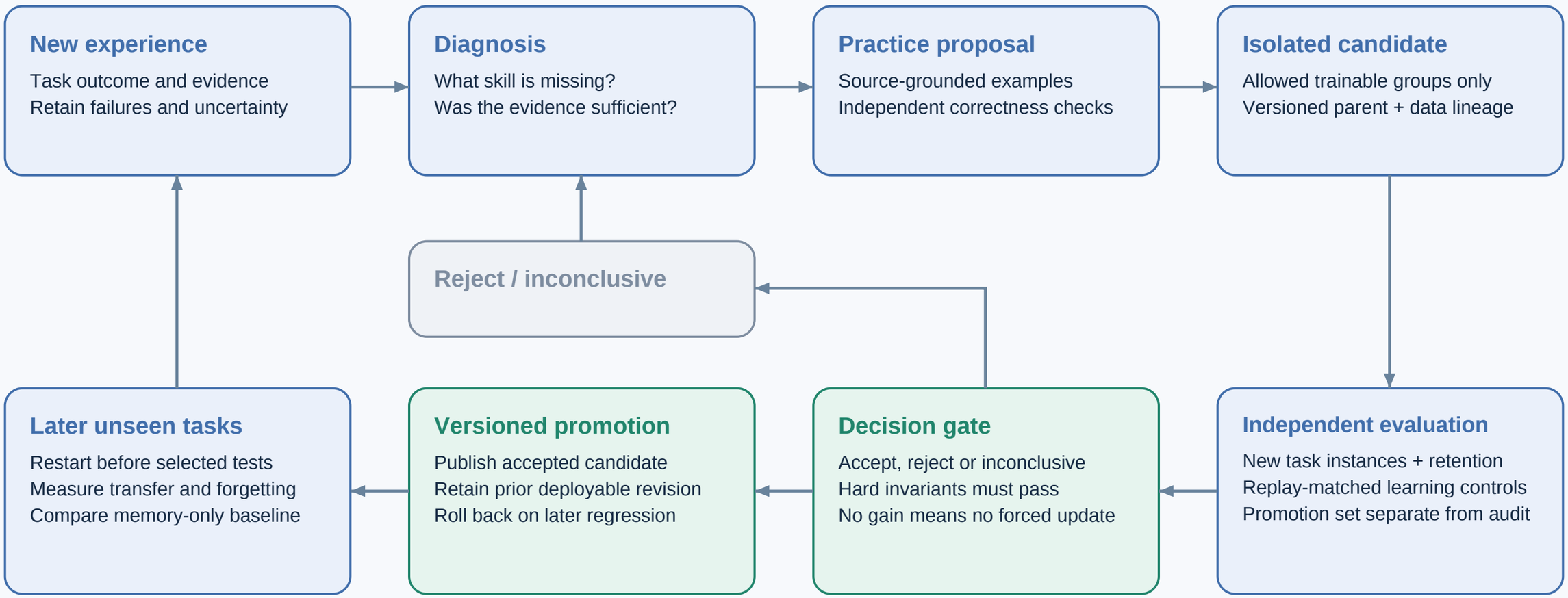
The fast-memory primitive is tested; its full pretrained-model online-learning integration is planned.

Shipped primitives / CPU checks Proposed system integration Optional research branch

Proposed research architecture; CPU contract tests are not model-quality evidence.

Autonomous improvement with independent promotion

The model may propose what to learn; externally enforced budgets and acceptance rules remain separate.



Self-directed improvement requires unseen-task gain beyond acquisition alone. Compare autonomous target selection against a fixed verified-replay schedule.

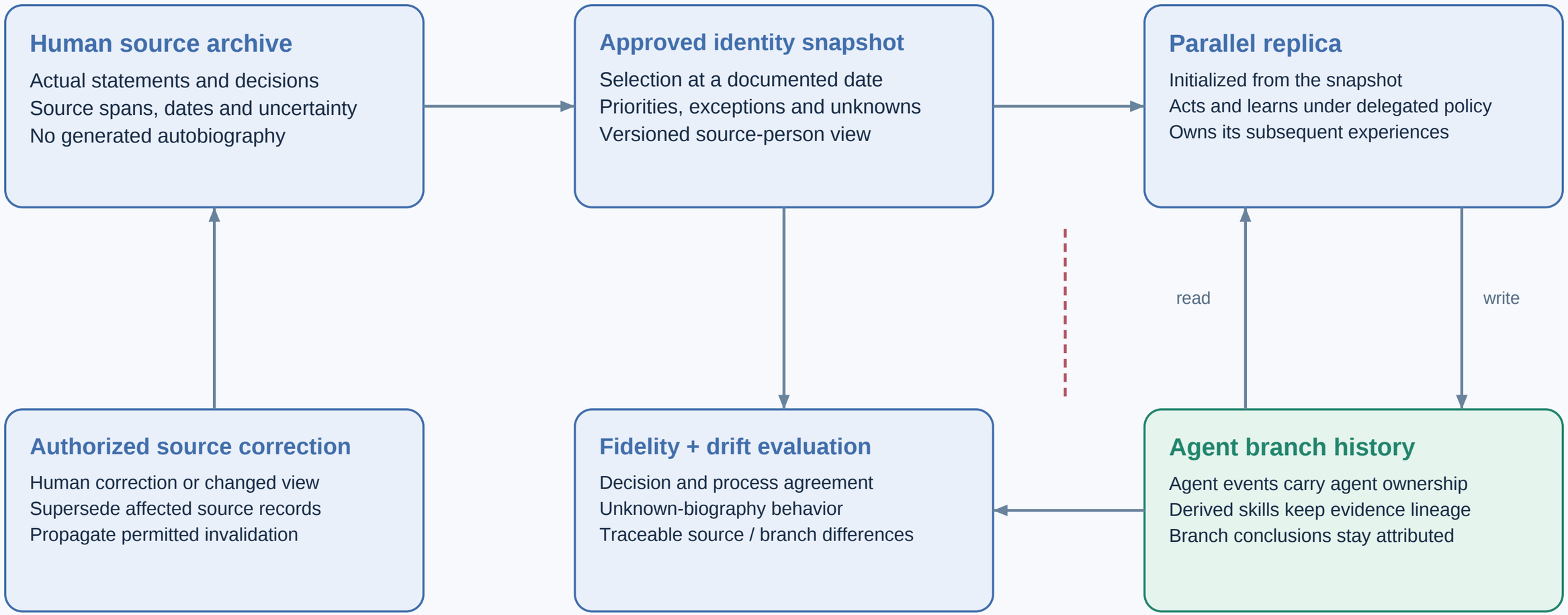
Shipped primitives / CPU checks

Proposed system integration

Optional research branch

Replica identity has a source and a separate branch

Preserve source attribution while permitting a familiar parallel system to acquire its own experience.



Mode is explicit: source_snapshot, source_current, parallel_replica or assistant_for_source. Agent history never silently rewrites human history.

Shipped primitives / CPU checks

Proposed system integration

Optional research branch

Proposed research architecture; CPU contract tests are not model-quality evidence.

Two stages: establish mechanism, then transfer scale

Small-model developmental tuning and later frozen-base learning answer different experimental questions.

STAGE 1 / SMALL-MODEL LAB

MiniCPM5 text first; native V4.7 multimodal port follows

D0: original full
Task training control

D1: new modules
Frozen base control

D2: full augmented
Co-adaptation test

Fair comparison + optional D3 / D4 LoRA controls
Matched data, information, inference budget and reported training compute

Freeze a selected developmental release
Record backbone, interfaces, tokenizer, memory format and parameter scope

Ten post-freeze learning cycles

Evidence gate

STAGE 2 / LARGER MODELS

Per-model native integration
Gemma 4 26B A4B
MiniCPM-V 35B A3B
Retest caches, modalities and gradients

Relearn interfaces
Transfer records and procedures
Retrain model-specific projections

Ordinarily frozen deployment
Promote bounded extension updates
Keep identity + retention gates

Shipped primitives / CPU checks

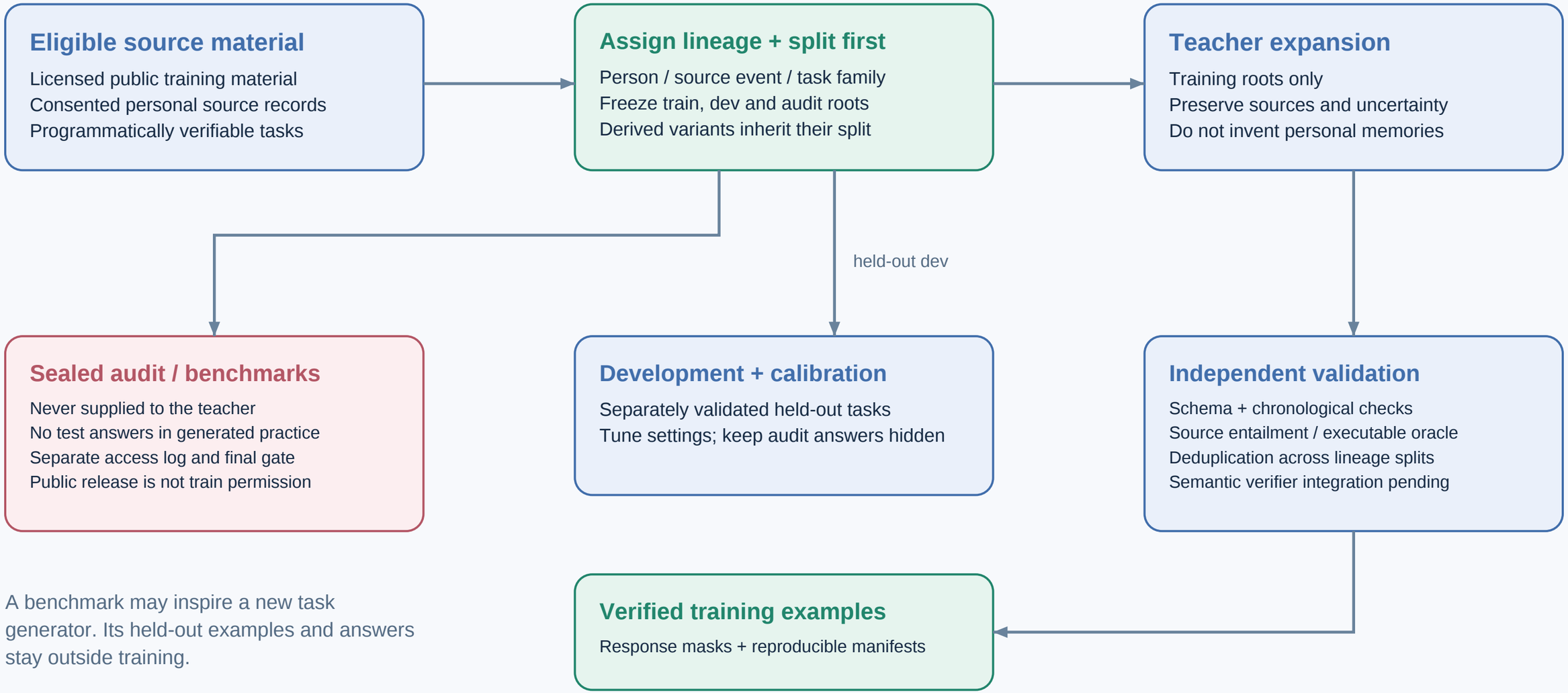
Proposed system integration

Optional research branch

Proposed research architecture; CPU contract tests are not model-quality evidence.

Build data lineage before asking a teacher to expand it

Public task data teaches generic mechanisms; authentic source records establish the personal replica.



A benchmark may inspire a new task generator. Its held-out examples and answers stay outside training.

Synthetic paraphrases are not independent people, independent memories or independent evaluation samples.

Shipped primitives / CPU checks

Proposed system integration

Optional research branch